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Micro-aeration for hydrogen sulfide reduction in full-scale anaerobic digesters with limited headspace: Performance and sulfide reduction kinetics

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ABSTRACT

A full-scale trial of micro-aeration was conducted at a municipal wastewater plant using two anaerobic digesters with limited headspace, one as a control and one as a test. A reduction of approximately 13 %–16 % in hydrogen sulfide biogas concentrations was observed in the test digester. Microbial community analyses were also conducted at steady-state conditions to determine differences between the microbial ecology of the test and the control digesters. Although sulfide removal typically occurs in the headspace of a micro-aerated digester, the data generated by this pilot study developed a new approach to quantify the contribution of liquid phase sulfide oxidizing microbes (SOM).

1. Introduction

Hydrogen sulfide (H_2S) is a common constituent of biogas produced during the anaerobic digestion (AD) of municipal wastewater sludge, and its concentration in biogas is dictated primarily by the sulfur content of the AD feed. Peu et al. (2012) reported H_2S concentrations in biogas as high as 19,000 parts per million (ppm). However, H_2S in municipal AD biogas typically averages 50–5000 ppm (Dumont, 2015). The presence of H_2S creates corrosive condensates that reduce the operational life of gas handling systems and damage gas pipelines and engines that receive the biogas in beneficial end-use projects.

Traditional biogas desulfurization techniques, such as granular activated carbon or wet scrubbing systems that use water or chemicals, require media replacement and disposal of the spent media. Micro-aeration, or the introduction of small quantities of air or oxygen into anaerobic digesters to facilitate the biological oxidation of sulfide by sulfide oxidizing microorganisms (SOM), is a promising, simple and economical biogas desulfurization technology (Fdz.-Polanco et al., 2009; Jeníček et al., 2017; Krayzelova et al., 2014; Nguyen and Khanal, 2018; Ramos et al., 2013). It has been successfully applied at full-scale municipal wastewater treatment plants with H_2S concentrations of 500–5000 ppm resulting in H_2S removal efficiencies of 85–99.5 % (Jeníček et al., 2017). Romero et al. (2020) presented results from three mesophilic full-scale micro-aeration applications, two in the US and one in Australia. The two anaerobic digesters in the US produced biogas at 4500–5000 m^3/d, with average H_2S concentrations of 1125–5000 ppm. Micro-aeration was applied by injecting air into the sludge recirculation lines, which achieved about 80 % reduction of H_2S. The Australian plant applied micro-aeration in the liquid phase to a mesophilic anaerobic digester with a relatively low biogas H_2S concentration of about 600 ppm. The micro-aeration performance for this system was unstable, with removal efficiencies averaging 40 %, occasionally reaching 90 %.

Most literature on the role of biogas residence time (RT) in biogas desulfurization was based on directly injecting air into the headspace of pilot units with biogas RTs of 1.4–8 h (Ramos et al., 2014; Kobayashi et al., 2012; Khoshnevisan et al., 2017). Removal efficiencies of 68 %–> 99 % were reported at these RTs; moreover, biogas RTs of five hours or more resulted in approximately 90 % removal efficiencies

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(Khoshnevisan et al., 2017). Ramos et al. (2014) evaluated the impact of micro-aeration in pilot scale digesters at a biogas RT of 6.8–8.6 h, which resulted in 97 %–100 % H_2S reduction. A full-scale study also evaluated air addition into the headspace of a mesophilic digester treating dairy cow manure (Kobayashi et al., 2012). At a biogas RT of 1.35–1.7 h and air addition at 1 % (v/v) of the biogas, the study achieved an average of 68 % H_2S removal efficiency for H_2S biogas concentrations of 2000–4000 ppm. Using biogas RTs reported by Khoshnevisan et al. (2017), Muller et al. (2022) established a first-order rate constant for H_2S removal in the range of 0.52 h^{-1}. However, data from full-scale plants on the impact of biogas RT with liquid phase injection of air/-oxygen are lacking and no information is available for digesters with limited headspace.

Not only is full-scale experience for digesters with limited headspace lacking, a recent review by Muller et al. (2022) also identified the need for better descriptions of SOM kinetics in the liquid phase. The available

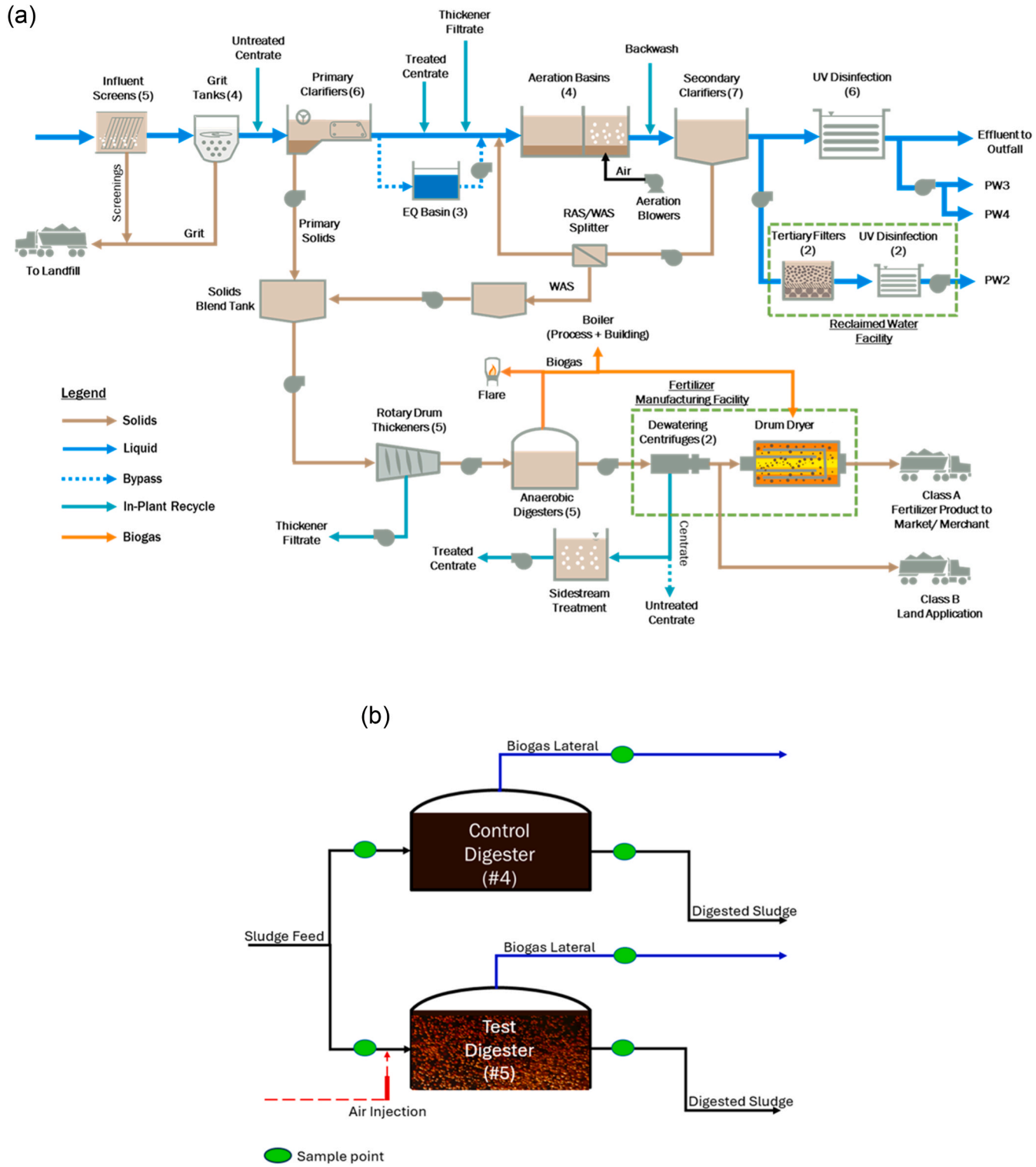


Fig. 1. a: Process flow diagram of the full-scale wastewater treatment plant at which the micro-aeration study was conducted. b: Schematic of the pilot set-up.

kinetic data from lab studies (Andreides et al., 2021; Nielsen and Vollertsen, 2021; Pokorna-Krayzelova, 2017) are derived from experimental designs using high dissolved oxygen concentrations that are not reflective of real-world conditions. Further exploration of SOM kinetics would better enable the delineation and quantification of liquid vs. gas phase desulfurization. This would enable the development of more accurate sulfur and COD mass balances as well as improved AD models.

Micro-aeration has potential as an economical biogas desulfurization method, but there are knowledge gaps regarding SOM kinetics and the effects of micro-aeration on AD performance metrics and the microbial community. There are inconsistent reports on improvements to AD performance metrics due to micro-aeration. Despite reported improvement in hydrolysis rates (Nguyen and Khanal, 2018), stabilization of volatile fatty acid (VFA) accumulation, and indications that the transformation of lactic, valeric, butyric, and propionic acids to acetate is enhanced in the presence of oxygen (Lim and Wang, 2013), the impact of micro-aeration on volatile solids reduction (VSR) and methane yield is contradictory. Jenicek et al. (2017) noted a 1.6 %–9.3 % decrease in the overall sludge volatile fraction with micro-aeration and average increases in biogas and methane production of 17 % and 16 %, respectively. In contrast, Diak et al. observed that micro-aeration of municipal primary sludge had no effect on the degradation of carbohydrates and VS reduction even though it enhanced the generation of soluble proteins and ammonia (Diak et al., 2013).

The objective of this study was to evaluate the biogas desulfurization efficiency at very short biogas RT, with air injection into the liquid rather than the headspace. To evaluate the efficacy of adopting micro-aeration in digesters with limited headspace, the study further assessed the contribution of liquid phase biological sulfide oxidation to the overall desulfurization. Sulfur mass balances were developed to assess the impact of micro-aeration on key performance indicators. Finally, the microbial community structures in micro-aerated and conventional digesters were compared to evaluate if micro-aeration resulted in greater SOM abundance.

2. Materials and methods

2.1. Evaluation of the impact of micro-aeration on desulfurization in pilot study

2.1.1. Plant description

The study was conducted at a wastewater treatment facility with an average dry weather flow capacity of 45 million gallons per day (mgd) (204,574 m^3/day); Fig. 1a shows the overall process flow diagram while Fig. 1b shows the schematic of the pilot setup. The ratio of particulate COD (pCOD): volatile suspended solids (VSS) in the co-thickened primary sludge and waste activated sludge were 1.36 and 1.62, respectively.

2.1.2. Micro-aeration setup and baseline operation

Two circular waffle-bottom anaerobic digesters, each 27 m in diameter, were used for this study: one as a control (D4) and one as the test digester (D5). Each digester has a working volume of 6965 m^3, a side water depth of approximately 12 m, and are mixed with draft tube mixing systems. The digesters are also equipped with submerged fixed covers that create a headspace volume of approximately 52 m^3, resulting in a biogas retention time of approximately15 min The micro-aeration study was divided into two phases: Phase 1: Baseline Operations, and Phase 2: Full-Scale Implementation. Phase 1 established a baseline of characterization for biogas and digestate without the addition of air. Biogas flow, influent sludge flow, and digestate flow rates were monitored continuously. Biogas composition was measured twice a day and pH was measured daily. Feed and digestate total and volatile solids were measured twice a week. Digestate alkalinity and volatile fatty acids (VFA) were measured twice a week.

During Phase 2, the existing plant service air system was used to supply air to the sludge recirculation line upstream of the test digester, through a ½" line with air flow meters with needle valves, capable of regulating airflow in the range of 20–200 cfh (0.6–5.7 m^3/h). Initial airflow rates were set at 90 scfh (2.5 m^3/h) for 49 days to ensure that oxygen levels in the gas did not exceed 0.2 %. Subsequently, airflow rates were doubled to 180 scfh (5 m^3/h) for 28 days and gas oxygen levels increased to the detection level of the gas meter. Then, the airflow rate was dropped to 140 scfh (3.9 m^3/h) for the remainder of the study (180 days). Since the RT in both digesters is about 40 days, only the samples corresponding to the operation at 3.9 m^3/h were analyzed for different chemical and microbial analyses as it spans more than three turnovers of the mean RT.

2.2. Offline batch test setup to evaluate SOM kinetics

Sulfur removal was investigated in two digestate samples taken from the control and test digesters a week before the termination of the microaeration trial. The digestate samples were subjected to air and N_2 injection to determine the biotic and abiotic (stripping) losses of dissolved sulfides, respectively. Once abiotic losses were determined, the rate of biological and chemical sulfide oxidation could be determined from the difference in rates between air and N_2 injections.

The experiments were conducted in two reactors, each with a volume of 1.7 L (12 cm internal diameter and 15.5 cm high), which were filled with 1.4 L digestate at 35 ℃. Both N_2 and air were injected to the reactor using a syringe pump (KD Scientific Inc., Holliston, Massachusetts, USA – LEGATO 100 Infuse Single model pump) equipped with a 5 mL Hamilton syringe and at a flow rate of 12 μL/min. This corresponds to a volumetric rate of 0.514 mL gas/L-h for 4 h, consistent with the full-scale micro-aeration rate of 3.9 m^3/h in the 6965 m^3 digester. While the digestate was introduced into the test reactor, N_2 gas at a flowrate of 0.3 m^3/h was injected for 4 min to purge the reactor of air and avoid sulfur oxidation during filling. The offgas was collected in a 1 L canister with an initial vacuum pressure of −28.7 inHg for final gas analysis. At the end of the experiment, the offgas was sent to a commercial laboratory (ALS Environmental Labs, London, ON) for measurement of H_2S. At time intervals 0, 30, 60, 90, 120, 180, and 240 min, samples were taken and measured for dissolved sulfide, sCOD, and VFA concentrations.

To assess the chemical oxidation of hydrogen sulfide, tests with 30 % hydrogen peroxide (H_2O_2—density 1.13 g/mL, 0.16 g O_2/mL) dosed at 100 times the rate of oxygen were also performed. Thus, to avoid oxygen limiting conditions, 30 % H_2O_2 was injected at a flowrate of 2.08 μL/min resulting in an injection of 100 times more oxygen with peroxide compared to air (0.33 mg O_2/min vs 0.0033 mg O_2/min).

2.3. Chemical analyses

Daily biogas production was measured in real time by the wastewater plant and composition wea measured using a BIOGAS 5000 m (Landtec QED, Dexter MI) throughout the pilot study. The digestate samples were tested for total sulfur (EPA 6010D, 2018), dissolved sulfur (Martin et al., 1994), and dissolved sulfates (EPA 375.2, 1993). The sCOD, VFA concentrations were analyzed using HACH methods (HACH, Loveland, Colorado). The total solids (Method 2540A), total suspended solids (Method 2540D), volatile solids (Method 2540B), and volatile suspended solids (Method 2540E) were analyzed following standard methods (American Public Health Association, 2023).

For offline batch test, the dissolved sulfide concentration was measured using HACH method (HACH, Loveland, Colorado). Sulfide TNT vials had a range of 0.1–2.0 mg S^{2-}/L, and therefore samples were diluted with deionized water prior to measurement.

2.4. Microbial analyses

Digestate samples from micro-aeration pilot study were shipped overnight on ice to the University of Michigan for microbial analyses.

Upon arrival, samples were centrifuged at 10,000 g for 10 min at 4 °C, the supernatants were decanted, and biomass pellets were stored at −80 ℃ until DNA extraction. DNA extraction was performed using DNeasy PowerSoil Pro Kit (QIAGEN, Venlo, The Netherlands) following the manufacturer's instructions with minor modifications, which included bead beating (Mini-Beadbeater-96, BioSpec Products, Bartlesville, OK, USA) three times for 2 min each using Lysing Matrix E tubes. DNA quality and quantity were determined via spectrophotometry (Nanodrop 1000, Thermo Fisher Scientific, Wilmington, DE, USA) and Qubit 2.0 Fluorometer (Invitrogen, Life Technologies, CA, USA). The DNA extracts were then submitted to the Host Microbiome Initiative (University of Michigan, Ann Arbor, MI, USA) for 16S rRNA gene sequencing via Illumina MiSeq (San Diego, CA, USA). Universal primers F515 and R806 targeting the V4 region of the 16S rRNA gene (Caporoso et al., 2011) were used for PCR amplification. The raw sequence data were submitted to NCBI under accession number PRJNA1149829. The sequences obtained were processed using DADA2 v.1.26 (Callahan et al., 2016). Analyses were performed using RStudio (version 2022.12.0.353).

2.5. Statistical analyses

The statistical analyses were conducted using Excel data tools packages. The *t*-tests were conducted at a 95 % confidence interval.

3. Results and discussion

3.1. Impact of micro-aeration on desulfurization

3.1.1. Overall performance of control and test digesters

Table 1 lists the overall averages and standard deviations of the main operational parameters for the control and test digesters during the baseline and micro-aeration at 3.9 m^3/h. The feed total solids concentration averaged 8.2 % (82 gTS/L) and the volatile solids averaged 7.31 % (73 gVS/L), representing an average volatile fraction of 89.2 %. The volatile fraction of raw primary sludge and WAS were 95 % and 84 %, respectively, while the ratio of particulate COD to VSS in the primary sludge and WAS were 1.36 and 1.62, respectively. Primary sludge accounted for 62.5 % of the combined primary sludge and WAS VS loading, as compared with 58 % during the micro-aeration study. The plant recycles digested sludge to ensure pumpability of the influent thickened combined sludge at 8.2 % solids, therefore the volatile solids reduction (VSR) efficiencies have been calculated on the basis of the influent primary and WAS solids listed in Table 1. It should be noted that despite the atypically high volatile fraction (76 % of the digested sludge), the VSR based on the influent average VS mass during the baseline study was about 54 %. The relatively high volatile fraction of the digestate is indicative of high influent non-biodegradable VSS, potentially from a fiber board manufacturing plant. During micro-aeration, VSR was 54.3 %, almost identical to the baseline condition, despite slightly higher influent flows.

Figs. 2a and 2b illustrate the temporal variation of biogas flow and specific methane yield (SMY in m^3/kg VS). It is evident that both biogas and SMY remained relatively stable during both the baseline study and the test period. The performance stability is corroborated by the low relative standard deviations of the various parameters depicted in Table 1. Using the aforementioned pCOD:VSS ratio of 1.62 for WAS and 1.36 for primary sludge, for the baseline study, the observed SMY theoretically corresponds to about 66 % VSR, which is greater than the observed VSR. However, during the micro-aeration study, the observed SMY reflects 57.7 % VSR, in close agreement with the 54.3 % reported above on VS mass basis.

3.1.2. Impact of micro-aeration on sulfide concentrations

Figs. 3a and 3b show the temporal variations of H_2S concentrations, and mass flows, respectively, while Fig. 3c shows the H_2S removal efficiency based on differences in concentrations and mass rates. It is evident from Fig. 3a that during the baseline study, there were no statistically significant differences in gaseous H_2S concentrations, with both digesters averaging about 2100 ppm. H_2S concentrations dropped by 20 % to about 1700 ppm in both digesters at the onset of the micro-aeration. It should be noted that as shown in Fig. 3b, even during the baseline study, the H_2S mass rate in the test digester were lower than the control due to the roughly 5 % lower gas production (Table 1) resulting from the slightly lower feed flow. The removal efficiencies in Fig. 3c clearly indicate that the H_2S concentrations in the biogas from the two digesters started to diverge with the initiation of micro-aeration. The average H_2S concentration in the test digester of 1470 ppm was 13 % lower than the control. On a mass basis, due to an 8 % lower biogas on average in the test digester relative to the control digester, the percent reduction in H_2S mass was 20 %.

. The differences between the control and test digester with respect to SMY ($p = 0.0997$), and methane production/mass of VS reduced ($p = 0.198$), were insignificant. Differences in digester VFA were statistically significant ($p = 0.0035$) with test digester averaging about 10 % higher VFA concentrations. The 13 % decrease in biogas H_2S between the test and the control digester discussed above was statistically significant ($p = 1.84E-69$). The difference between the control digester average methane content of 57 %, and 56 % in the control and test digesters was statistically significant ($p = 1.37E-12$).

Table 1
Average operating conditions and digester performance during Phase I (Baseline) and Phase II (Micro-aeration study).

Parameter	Baseline (no aeration)		Micro-aeration	
	Control	Test	Control	Test
Feed sludge, gpd	48,000± 6200	46,600± 5600	51,200± 6900	49,900± 6400
Feed sludge, %TS	8.2± 0.5		8.2± 0.6	
Feed sludge, % volatile fraction	89.2± 0.7		89.2± 1.0	
HRT, d	40.4± 5.3	41.5± 4.9	37.8± 5.1	38.8± 5.1
Primary solids to both dig., kg/d	12,773± 773		12,773± 1591	
WAS solids to both dig., kg/d	8636± 1182		10,500± 1136	
Digestate, %TS	3.28± 0.1	3.25± 0.1	3.22± 0.1	3.2± 0.1
Digestate, volatile fraction, %	75.6± 1.1	75.4± 0.8	80.2± 0.4	75.1± 1.5
Average VSR, %	53.6	53.8	51.2	54.3
Digestate volatile fatty acids (VFA), mg /L as acetic acid	168± 56	153± 45	70.0± 12.2	78.3± 18.6
Biogas flow, 1000 m^3/d	5.9± 0.5	5.6± 0.4	5.9± 0.4	5.4± 0.4
Methane content of biogas, %	58.6± 0.8	58.4± 0.8	57.0± 0.8	55.6± 2.0
Biogas residence time, min (calculated)	13.6± 1.0	14.2± 1.0	14.9± 1.3	16.1± 1.5
Methane production, 1000 m^3/d	3.5± 0.3	3.3± 0.3	3.4± 0.2	3.0± 0.2
Specific methane yield, m^3/kg VS	0.43± 0.04	0.42± 0.04	0.45± 0.11	0.44± 0.11
H_2S, ppm	2080± 140	2100± 130	1690± 94	1470± 73
H_2S, kg/d	16± 0.7	16± 0.8	13.2± 0.7	10.6± 0.6

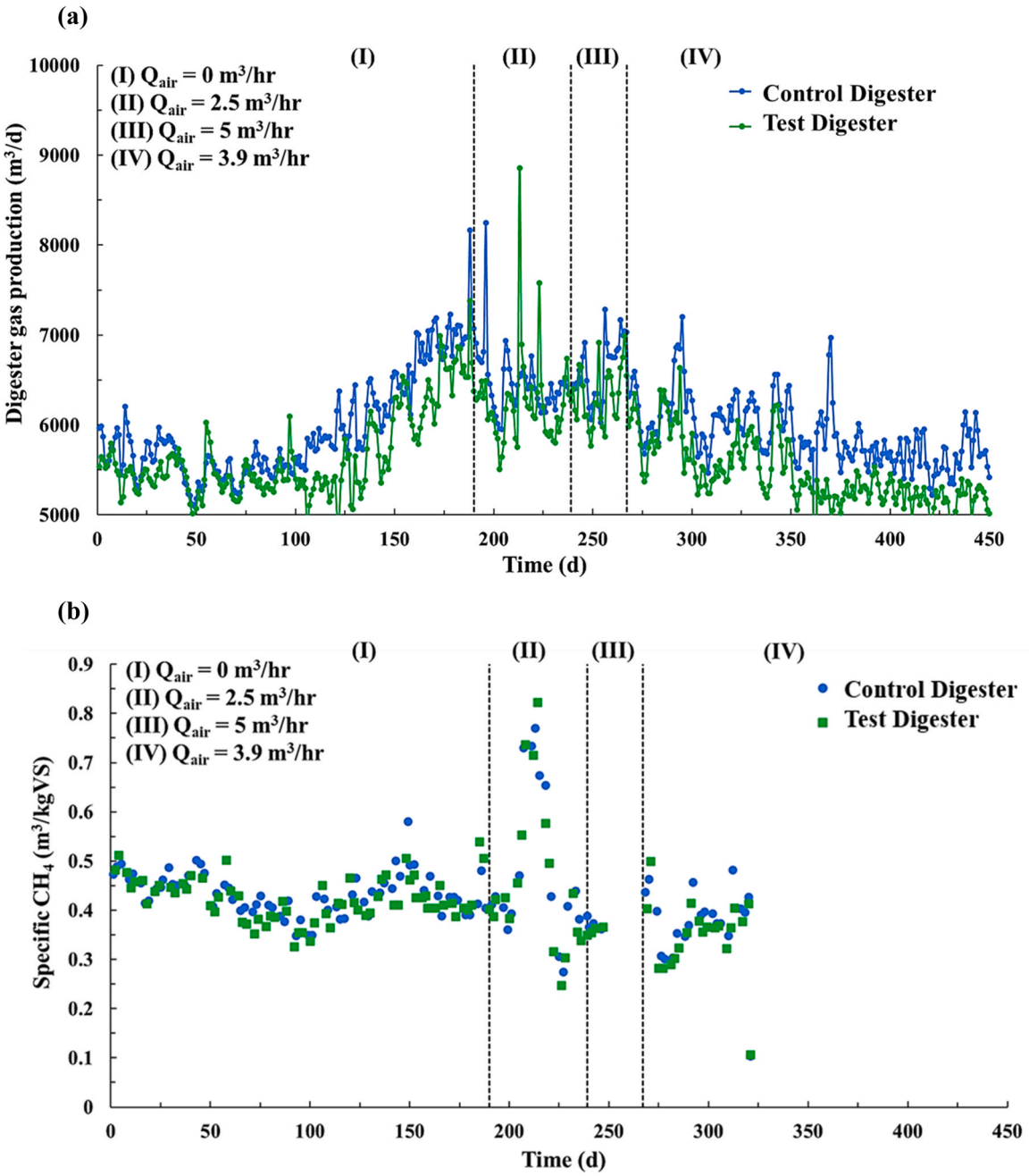


Fig. 2. a and 2b. Temporal Variations of Biogas Production and Methane Yields in the Control (D4) and Test (D5) Digesters.

Oxygen in the control digester biogas was negligible but averaged 0.5 %$\pm$ 0.01 % in the test digester. Using the correlation between % O_2 concentration in the biogas reported by Muller et al. (2022) being equal to 0.22 x volumetric percentage of airflow/biogas flow, the estimated average oxygen in the test digester biogas is 0.4 %, in close agreement with the observed average. The mass ratio of oxygen supplied: H_2S removed was 10.8$\pm$ 3.1, much higher than the stoichiometric requirement of 0.5 kg O_2/kg S. Giordano et al. reported that in full-scale systems with micro-aeration applied to the digester headspace, typically the air supplied should be 4 times the theoretical requirements (Giordano, et al., 2019). The considerable increase in demand above theoretical requirements is attributable to myriad other processes in the digester (facultative organisms scavenging dissolved oxygen, for example) which can consume up to 59 % of the oxygen supplied. Although delineating the oxygen transfer efficiency in the test digester was beyond the scope of this work, considering the 12 m-deep digester, oxygen transfer efficiency to the liquid is expected to be very high, thus resulting in low oxygen in the digester headspace, rationalizing the high O_2: H_2S ratio.

It should be noted that based on the VSR reduction efficiencies of about 54 %, the SMY should be around 0.36–0.38 m^3/kg VS in, as indicated by the averages reported in Table 2, and thus values above that range are not realistic and only reflective of the heterogeneity of the feed and lack of representativeness of the grab sampling approach used. Figs. 4a and 4b show a downward slope of the SMY with gaseous H_2S concentrations in the range of 1000–1500 ppm, suggesting potential inhibition due to H_2S. Similarly, within the observed SMY range, the SMY appears to increase with decreasing VFAs. Sulfide toxicity to methanogens have been reported to follow a non-competitive inhibition model (Paula Jr. and Foresti, 2009) which is the default inhibition model in ADM1 (Barrera et al., 2015). The inhibition constant (K_I), corresponding to 50 % inhibition has been reported to vary from 10 mg

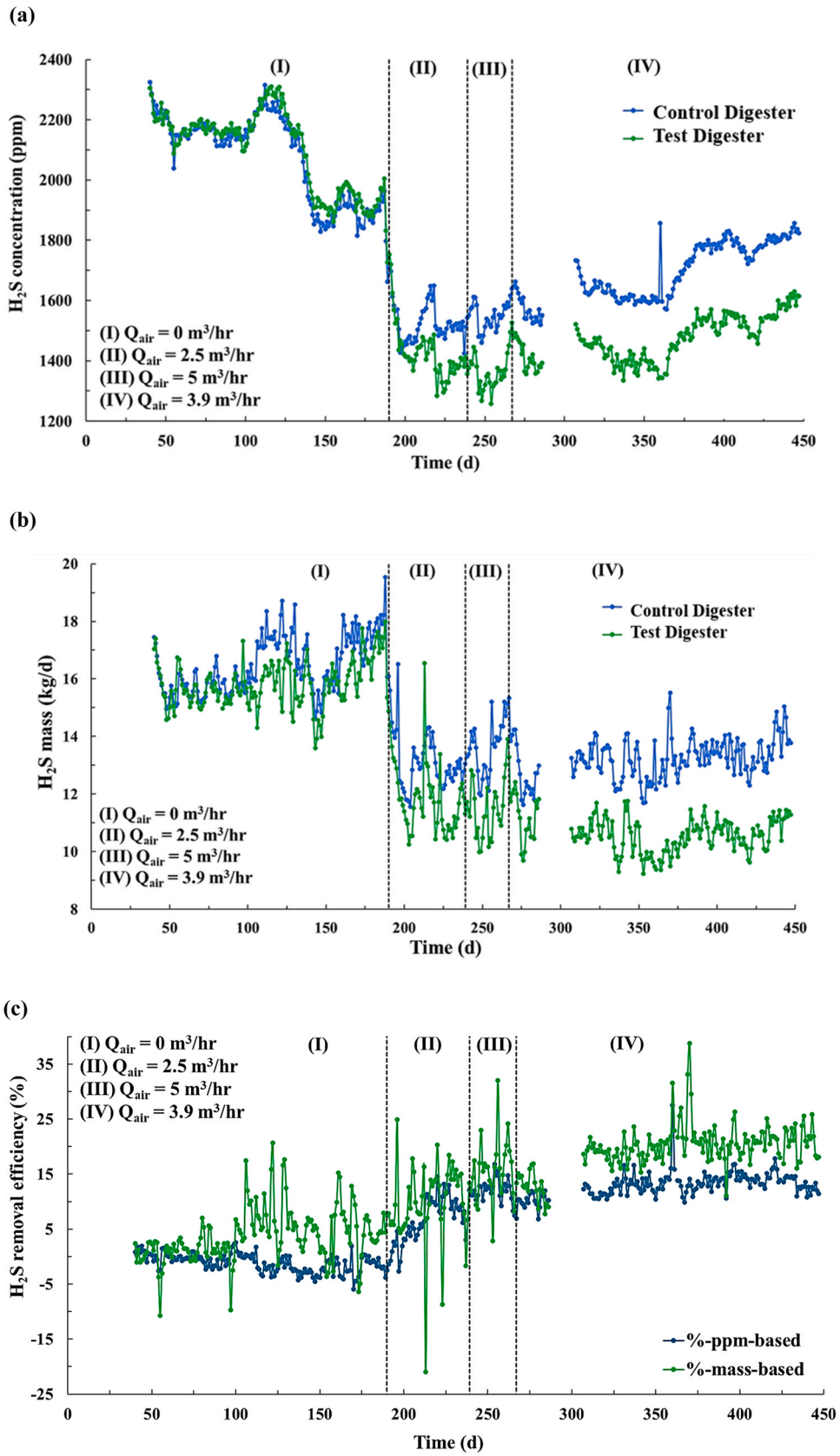


Fig. 3. Temporal Variations of (a) H_2S concentrations, (b) mass rates, and (c) sulfide removal (%) in the Control and Test Digesters.

Table 2
Sulfur mass balance in the micro-aeration study.

Parameter	Units	Control digester	Test digester
Total solids to both digesters	kg/d	**23,273**	
Total sulfur to both digesters	kg/d	**209**	
Influent flow	gal/d	**51,200**	**49,900**
Total solids concentration	mg/L	**32,200**	**32,000**
Particulate sulfur	kg/d	**60**	**52**
Soluble sulfur	kg/d	**11**	**12**
Gaseous H_2S	kg/d	**13.2**	**10.6**
Total effluent sulfur	kg/d	**85**	**75**
Sulfur mass balance based on flow	%	**77**	

H_2S-S/L (Robels et al., 2020) to 320 mg H_2S-S/L (Paula Jr. and Foresti, 2009). At 37 °C, the dimensionless Henry's constant for H_2S is 0.5 (Metcalf et al., 2013), and a thus a gaseous concentration of 1 ppm corresponds to a concentration of 1.34 mg/m^3 air (Metcalf et al., 2013). Thus, 1 mg/L of dissolved H_2S is equivalent to 373 ppm. At the pH of 7.3 at 37 °C, the pKa1 for H_2S is 6.75 (Migidisov et al., 2002), and thus the unionized fraction of dissolved sulfides is 22 %. Hence, the gaseous H_2S concentrations of about 1500 ppm would correspond to 0.88 mg H_2S-S/L, and thus the reduction in specific methane yield ($1-(K_I/(K_I+S))$, based on the two extremes of K_I would range from 0.3 % to 8 % only. It should be noted that the relative standard deviation of the SMY reported in Table 1 is about 16 %, which would mask the reported SMY reduction.

3.1.3. Evaluation of sulfur species

Muller et al. (2022) identified the lack of reliable sulfur mass balances, both in full-scale and pilot-scale applications. For plants that use iron to precipitate phosphorous, the sulfur content of digestate on dry mass basis is typically higher than the raw sludge (Dewil et al., 2009), which combined with the difficulty of quantifying sulfur precipitated in biogas headspace and piping, makes the closure of the sulfur mass balance very challenging. Dewil et al. assessed the fate of sulfur in four full-scale wastewater treatment plants in Belgium and observed that sulfur concentrations in biosolids (as a percent of dry solids) increased significantly after digestion at facilities that use iron to precipitate phosphorous, as iron bound to the released sulfides. For plants that do not use iron, there was essentially no change in sulfur content after digestion.

The facility at which this study was conducted does not add iron for phosphorus recovery. Seven samples from each of the two digesters (control-D4, and test-D5) were collected at the end of the micro-aeration study, the longest steady state operating period, and analyzed for total sulfur. The sulfur content as a percent of total solids averaged 0.86 %$\pm$ 0.12 % in D5 and 0.95 %$\pm$ 0.29 % in D4. Similarly, the average total dissolved sulfur in D4 and D5 were 64$\pm$ 40, and 57$\pm$ 59 mgS/L, respectively. The average dissolved sulfides in D4 and D5 were 11.6$\pm$ 10.3, and 9.7$\pm$ 11 mgS/L, respectively. Table 2 presents the sulfur mass balance to the digesters based on the average parameters reported in Table 1, with a mass balance closure of 77 % for the digestion system. At the pH of 7.3 at 37 °C, the pK_{a1} for H_2S is 6.75 (Migidisov et al., 2002). An absence of sulfates in the reduced anaerobic environment means the 22 % un-ionized fraction of dissolved sulfides corresponds to 2.4 and 2.7 kg/d, in the control and test digesters, respectively. Therefore, in the control digester, the gas phase H_2S mass rate was 5.5 times that in the liquid phase, as compared to four-fold in the test digester. This clearly attests to the significant effect of stripping and volatilization of H_2S from the liquid, implying that effective biological desulfurization predominantly needs to occur in the gas phase. Furthermore, given the slow kinetics of SOM in the liquid phase under micro-aerobic conditions, as discussed later, the effect of liquid phase biological oxidation is bound to be of limited efficacy. The precipitation of H_2S to sulfur in the liquid phase should result in a proportional increase in the sulfur content of digestate. However, the total sulfur, as a percent of total solids, in the test digestate was lower than the control despite similar total solids (Table 1), which indicates that the sulfur precipitation appears to have occurred predominantly in the headspace. The 23 % discrepancy between the effluent and influent total sulfur mass rates could be due to a combination of factors, such as loss of H_2S in feed and recirculation pumps, sulfur deposited on the digester walls, cover and gas pipes (as shown in SI Figure SI-3), as well as variations in raw sludge quality. Paing et al. conducted a detailed evaluation of an anerobic lagoon over a one-year period to both estimate H_2S emissions and develop the sulfur mass balance, and concluded that despite the extensive sampling and monitoring, about 30 % of the influent total sulfur could not be accounted for (Paing, et al., 2003). Ruan et al. studied the speciation of various forms of sulfur in lab-scale micro-aerated anaerobic digesters loaded with scrap iron for H_2S control and accounted not only for soluble and particulate sulfur but also sulfur precipitated on the digester walls, reported a sulfur mass balance closure of 90 %, with sulfur deposits on the digester walls contributing about 20 % to the sulfur mass in the system (Ruan et al., 2017). The variability of sulfur emissions from various unit processes was demonstrated by Forouzanmehr et al. during a two-week sampling campaign of a 620,000 population equivalent IFAS-based wastewater treatment plant in South France. The study reported that while the relative standard deviation of sulfur mass rates in sludges and liquids was mostly around 20 %, it increased to 50 % in gaseous emissions (Forouzanmehr, et al., 2021).

3.1.4. Impact of micro-aeration on microbial communities

The dominant amplicon sequence variants (ASVs) present at a relative abundance higher than 2 % in the two digesters were related to the genera *Treponema*, *Smithella*, *Rikenellaceae RC9 gut group*, *Fastidiosipila*, *Blvii28 wastewater-sludge group* (Acetobacteroides) (Fig. 5). Furthermore, *Methanolinea* and *Methanothrix* were the most abundant hydrogenotrophic methanogenic and acetoclastic methanogenic genera, respectively.

As discussed above and elaborated upon later, no H_2S removal was observed by the biomass in the liquid phase of the digesters, suggesting that the contribution of SOM in the digestate to the overall H_2S removal was insignificant. Consistent with this, no known SOM genera were observed to be present at a relative abundance higher than 2 % by the 16S rRNA gene analysis of the biomass samples obtained from the digestate. The SOM belonging to the genera *Acidithiobacillus*, *Acinetobacter*, *Rhodococcus*, *Arcobacter*, *Sulfuricurvum*, *Halothiobacillus*, *Sulfurimonas*, *Thiomonas*, are typically found in anaerobic bioreactorss subjected to micro-aeration. While there are a few studies that analyzed sulfide oxidation in the bulk solution, validation of sulfide oxidation based on the microbial communities was limiting (de Arespacochaga et al., 2014), as the relative abundances of SOM are very low. A micro-aeration study on lab-scale digesters treating chicken manure with a 40 d SRT achieved a 99.7 % reduction in biogas H_2S through air injection into the liquid phase, with SOB (*Sulfuricurvum, Thiobacillus, Acinetobacter, and Psedomonas)* present at low relative abundances (0.1 %–0.6 %) in (Song et al., 2020). Similarly, in this study, *Acinetobacter, Arcobacter*, and *Pseudomonas* were observed at low relative abundances ranging from 0.03 % to 0.25 %. While oxygen affinity coefficients of the SOM have been observed lower than the air flow rates of this study, the microbial analysis and physicochemical tests suggest that there might be a possibility that the air supplied to the reactor was not utilized by the microorganisms in the liquid and escaped back to the headspace due to poor mixing conditions.

3.2. Kinetics of liquid-phase desulfurization in offline batch tests

The main objective of the offline batch testing was to assess the feasibility of biological sulfide oxidation with micro-aeration, under limiting dissolved oxygen conditions (below detection limits). The other objective was to assess whether micro-aeration improved hydrolysis and acidification in the digestate.

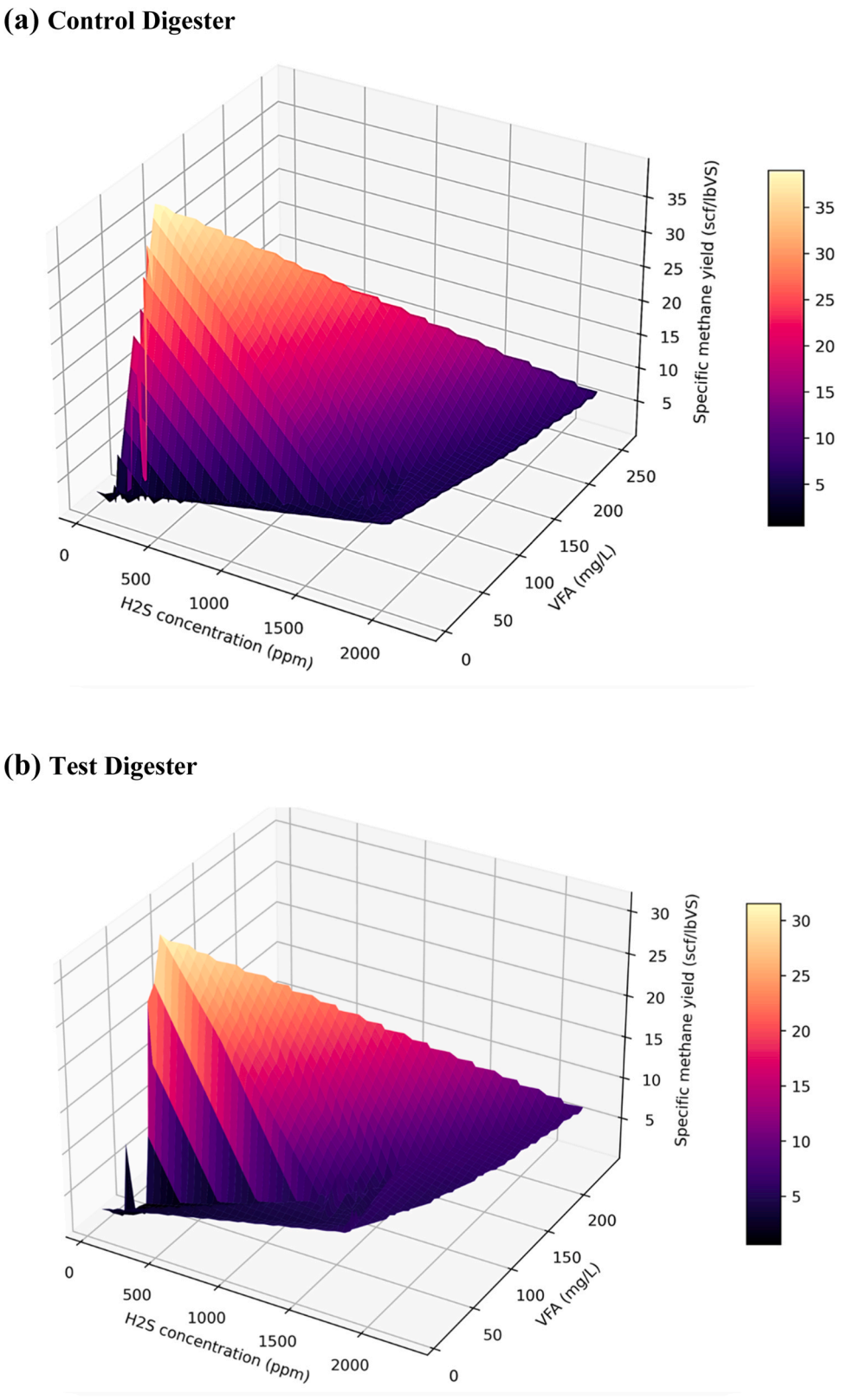


Fig. 4. a and 4b. Variation of Specific Methane Yield with Gaseous H_2S and Volatile Fatty Acids—(a)-Control Digester (Digester 4), and (b)-Test Digester (Digester 5).

3.2.1. Sulfide oxidation in batch tests

Air, nitrogen, and hydrogen peroxide were introduced into the 1.4 L reactor that contained control digester sludge at rates of 12,12, and 2.8 μL/min, respectively over a course of 4 hours.

Figs. 6a and 6b illustrate the effect of different gas injections on the total dissolved sulfides over time in the lab control and test digesters, respectively. The initial sulfide concentration of 25 mg/L decreased linearly to 15 mg/L by the end of the test using air as the aeration

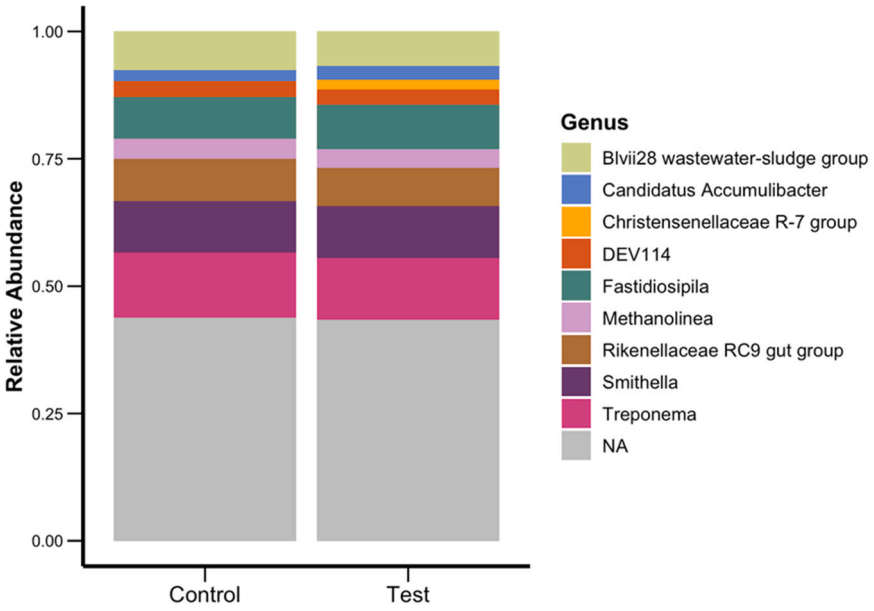


Fig. 5. Relative abundances (a) microbial (both bacterial and archaeal) communities and archaeal (right)of dominant ASVs communities in control and micro-aerated digesters at the Genus level. NA: Not Available; refers to the genera of microorganisms whose relative abundances were low and below the detection limit (2 %).

medium. Similar trend was observed with N_2 injections where the initial sulfide concentrations of 24 mg/L decreased to 16 mg/L over 4 hours. As evident from Fig. 5, under both micro-aeration with air or N_2, the dissolved sulfides decreased linearly following zero-order kinetics, with specific rates of 0.042, and 0.039 mg/L-min. Using H_2O_2 at 100 times the oxygen flow rate (as air), the rate increased by 33 % to 0.0565 mg/L-min. In summary, using sludge from the full-scale control digester, total dissolved sulfide reductions of 40 %, 34 %, and 61 % were realized for air, N_2, and H_2O_2 dosing, respectively.

The same pattern was also observed for batch tests using digested sludge from D5, the full-scale test digester. As illustrated in Fig. 6b, with air, initial sulfide concentrations decreased linearly from 21 mg/L to 11 mg/L after 4 hrs; similar results were seen with N_2 introduction where sulfide levels dropped from 16 mg/L to 9 mg/L, corresponding to rates of 0.035 and 0.029 mg/L-min, respectively. The rate with H_2O_2 increased by 31 % to 0.046 mg/L-min. These align with the results from the batch tests using control digester. It is worth noting that the total amount of oxygen injected over the 4-hr test would stoichiometrically result in oxidation of maximum of 0.92 mg S/L.

Interestingly, sparging with air in both batch tests removed an incremental 3 mgS/L relative to N_2, which indicates that neither chemical nor biological sulfide oxidation but rather stripping accounted for the increased removal of sulfides with air. Using the typical H_2O_2 dose required to oxidize sulfides to sulfur of 2.4 mg H_2O_2/mgS (Fernando and Peters, 1988), the H_2O_2 added during the 4-hr test would theoretically oxidize 41 mgS/L, however, incremental sulfide removal with H_2O_2 relative to N_2 was 6 mgS/L in both batch studies. Considering the strong oxidation potential of organics with H_2O_2, it could be assumed that most of the peroxide was not used for sulfide oxidation. However, this assumption cannot be verified due to the lack of relevant data.

Buisman et al. who examined chemical oxidation of H_2S levels of 5–319 mg/L using dissolved oxygen concentrations of 0.1–8.5 mg/L determined that the rate of oxidation can be described by $H2S^{0.41} \times O2^{0.39}$ (Buisman, et al., 1990). Given that dissolved oxygen concentrations during the offline batch tests are expected to be very close in the two systems, the rate of H_2S chemical oxidation would be proportional to H_2S concentration. As evident from Figs. 6a and 6b, average dissolved sulfide concentrations during the 4-h test in the D4 batch was 20 mg/L as compared with 15 mg/L in the D5 batch, with identical pHs of 7.3. Thus, the expected ratio of oxidation rate in the test digester to the control digester is 81 %, very close to the observed 82 %. Thus, notwithstanding the stoichiometric oxygen limitations, the slightly lower aerobic sulfide removal rate in the test digester compared to the control suggests that sulfides were chemically oxidized rather biologically oxidized.

The contribution of liquid phase sulfur oxidation to the overall H_2S removal will likely be very low due to the stripping of oxygen by the biogas production in the digester, however testing this in the offline batch test was not feasible. Comparing the immediate sulfide reduction in the control offline batch digester to the full-scale micro-aerated digester, which took approximately 30 days to show sulfide reduction, further indicates that the relative contribution of microbial liquid phase desulfurization was minimal in the batch tests.

The average biogas RT in the headspace of the full-scale test digester was approximately 16 minutes. Based on the first-order rate of 0.52 h^{-1}, derived from pilot and full-scale studies with headspace micro-aeration by Muller et al. (2022), the biogas RT of 16 minutes would correspond to an estimated concentration-based removal efficiency of 13 %, matching the observed 13 %.

Typically, upper limits of H_2S concentrations for biogas utilization vary depending on biogas use with limits varying from < 100 ppmv for combined heat and power engines (CHP) and < 4 ppmv for natural gas upgrade (Choudhury et al., 2019). Thus, based on the observed 2100 ppmv without micro-aeration in this study, and the aforementioned rate constant of 0.52 h^{-1}, the required biogas residence times would be 5.8 and 12.0 hours for CHP and natural gas upgrade, respectively. Thus, in anaerobic digesters with limited headspace and/or floating covers biological desulfurization of biogas in the headspace is insufficient to meet the requirements, and therefore dedicated scrubbers, bioscrubbers, adsorbers, and other clean-up technologies are required downstream. Although the biofilm surface area in the headspace can be increased by the addition of carrier media, such measures are unlikely to improve performance by the factor of 20–50 needed for necessary biogas quality upgrade.

3.2.2. *Hydrolysis and acidification*

Table 3 summarizes the measured VFA and sCOD during the 4-h offline batch tests for both digestate. Micro-aeration did not improve

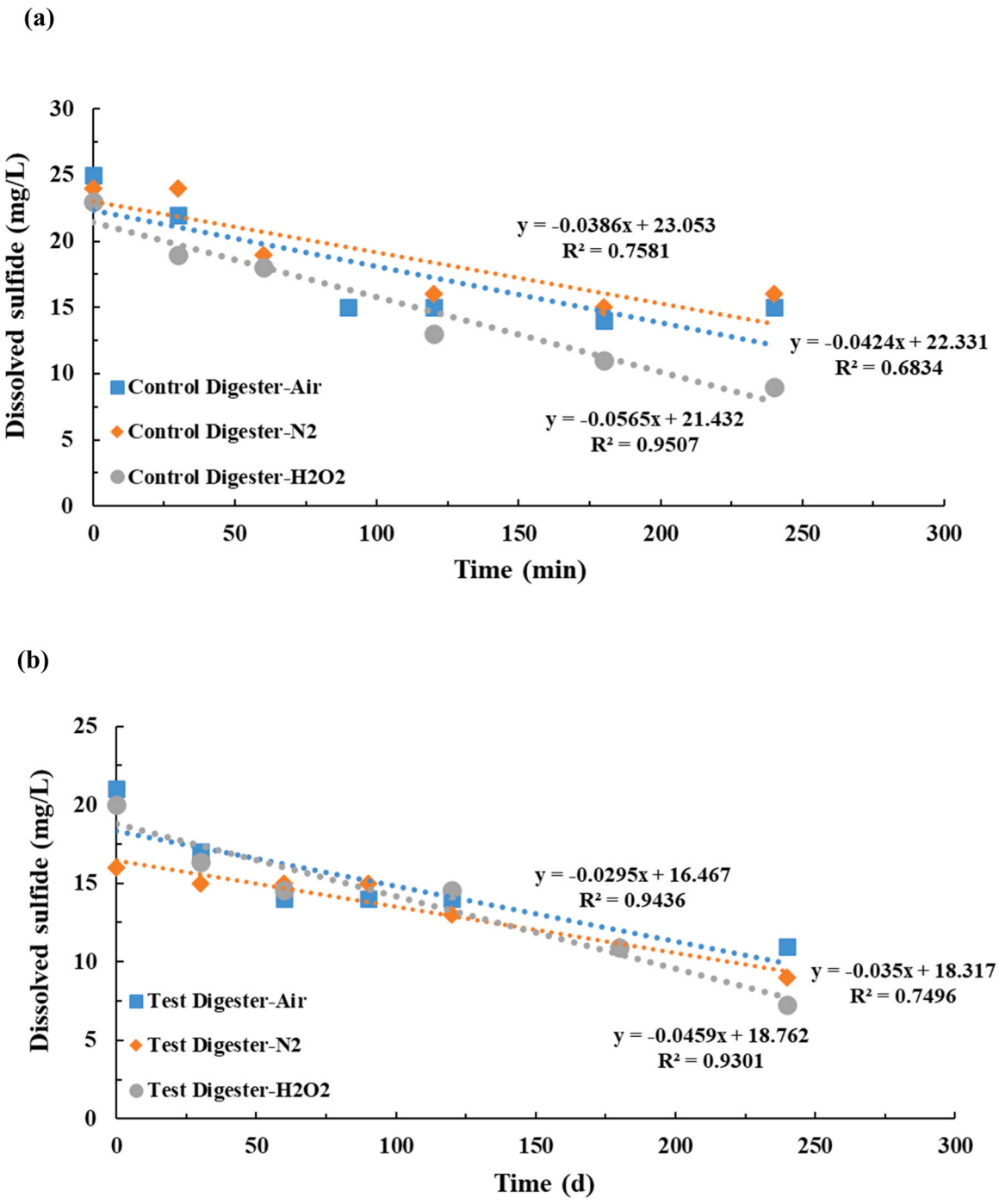


Fig. 6. Dissolved sulfide concentration during offline micro-aeration in the lab scale control digester (a) and test digester (b).

both hydrolysis and acidification within the 4-h test period, which was expected because of the already-stabilized nature of the test digestate and absence of raw sludge. However, the sCOD and VFAs were 8 %, and 58 % higher, respectively, in D5 (test) than D4 (control). The higher sCOD is indicative of higher solubilization and hydrolysis over time in the full-scale digesters subjected to micro-aeration.

The offline batch testing revealed several limitations that must be addressed. The stripping of the H_2S by biogas production in continuous-flow digesters will decrease the dissolved H_2S and slow the rate of both chemical and biological oxidation of sulfides in the liquid phase, despite being advantageous to biotreatment in the headspace. To simulate the conditions in the digester, with respect to both the generation of dissolved sulfides and their stripping by the produced biogas, the batch tests should involve addition of the feed sludge, as a control. Due to differences in oxygen transfer efficiencies between the full-scale digester and the offline batch digester, air injection rates must be carefully determined to ensure that the same redox potential, not the same volumetric air-to-biogas flows is maintained as the full-scale. Maintaining the same redox potential ensures that the chemical oxidation of sulfides by various species is properly simulated in the offline test. The presence of even trace concentrations of iron in the digestate may bind with sulfides interfering the results and thus, total and dissolved iron species need to be monitored during the test. Offline tests should be done, using both the digestate and biofilms attached on digester walls. While this might be easy for bench-scale laboratory digesters, obtaining biofilms from full-scale facilities may be challenging.

Table 3
sCOD and VFA variations in the offline batch tests.

	D4 (Test) Digestate				D5 (Control) Digestate			
	Air		N_2		Air		N_2	
Time (min)	VFA (mg/L)	sCOD (mg/L)	VFA (mg/L)	sCOD (mg/L)	VFA (mg/L)	sCOD (mg/L)	VFA (mg/L)	sCOD (mg/L)
0	260	1694	274	1804	368	1835	361	1875
30	234	1518	256	1958	402	1910	417	1855
60	234	2090	234	1716	376	1830	351	1910
90	246	1540	294	1958	390	1995	435	1895
120	282	1628	272	1408	383	1955	406	1970
180	236	1782	242	1650	396	1850	372	1905
240	240	1738	270	1716	403	1830	393	1795
Average (mg/L)	247 ± 18	1713 ± 193	263 ± 21	1744 ± 191	388 ± 18	1886 ± 68	390 ± 31	1886 ± 54

4. Conclusions

This work demonstrated the feasibility of using micro-aeration for biogas desulfurization in digesters with very limited headspace by injecting air into the liquid phase as opposed to the digester headspace, achieving, about 13 % reduction at a biogas residence time of only 16 minutes. In digesters with limited headspace, although micro-aeration cannot achieve the low H_2S for biogas utilization in CHP and natural gas upgrade, it can be readily applied to reduce loadings to downstream desulfurization systems economically. Micro-aeration was beneficial for hydrolysis, as evidenced by the higher soluble COD in the test digester.

Offline batch testing indicated that in the absence of biogas production, the kinetics of dissolved sulfides reduction with N_2, air, and H_2O_2 followed zero-order, with rate constants varying from 0.029 to 0.056 mg/L-min, with chemical oxidation as the primary removal mechanism and no major contribution of SOM to dissolved sulfides reduction. The applicability of the empirical first-order H_2S removal rate of 0.52 d^{-1}, developed by Muller et al. (2022) based on full-scale and pilot-scale headspace micro-aeration, was evidenced by the identical theoretical and experimental H_2S reduction efficiencies of 13 %.

Future research should focus on the optimization of micro-aeration in limited headspace digesters by the addition of biofilm carrier media in the headspace, evaluation of headspace kinetics particularly at low H_2S concentrations that may restrict biofilm growth due to diffusional limitations, and delineation of appropriate scaling criteria. The determination of the kinetics of dissolved sulfide oxidizing bacteria in the lab, under oxygen-limited conditions similar to full-scale digesters remains a significant challenge, complicated not only by the chemistry of sulfide precipitation, but also by the need to emulate the same mass transfer conditions of the full-scale digester. Insight into the correlation between the relative abundances of various sulfur oxidizing bacteria and biological sulfur oxidation is lacking.

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CRediT authorship contribution statement

Bell Kati: Resources. **Shahbaz Parisa:** Writing – review & editing, Resources. **Pierce Laurie:** Resources. **Summers Amanda:** Writing – review & editing, Resources. **Skerlos Steven:** Supervision. **Nakhla George:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis. **Bronstad Embrey:** Writing – review & editing. **Khadir Ali:** Writing – original draft, Investigation, Methodology. **Guevarra Karla:** Writing – review & editing, Resources. **Muller Christopher:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Raskin Lutgarde:** Writing – review & editing, Supervision, Methodology. **Karki Renisha:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.psep.2025.106911.

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